

Angle Rules and Polygons

GCSE Maths · Practice Questions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Two angles lie on a straight line. One is 115° .
Work out the other angle.

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2 [2 marks]

Three angles meet at a point. Two of them are 90° and 130° .
Work out the third angle.

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3 [2 marks]

Two straight lines cross. One of the angles formed is 72° .
(a) Write down the angle vertically opposite it.
(b) Work out one of the other two angles.

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4 [2 marks]

A triangle has angles of 43° and 90° .
Work out the third angle.

.....

5 [3 marks]

An isosceles triangle has an apex angle of 40° .
Work out the size of each of the two base angles.

.....

6

[3 marks]

A quadrilateral has angles of 85° , 100° , 120° and x° .

Work out the value of x .

7

[3 marks]

(a) Work out the size of one exterior angle of a regular pentagon.

(b) Work out the size of one interior angle of a regular pentagon.

8

[3 marks]

Two parallel lines are crossed by a straight line. One of the angles formed at the first crossing is 118° .

(a) Write down the size of the corresponding angle at the second crossing, with a reason.

(b) Work out the size of the co-interior angle, with a reason.

9

[4 marks]

(a) Work out the sum of the interior angles of a hexagon.

(b) Five of the angles of a hexagon are 100° , 120° , 130° , 140° and 110° . Work out the sixth angle.

10

[4 marks]

A regular polygon has an exterior angle of 24° .

(a) How many sides does it have?

(b) Work out one interior angle.

(c) Work out the sum of its interior angles.

1 [2 marks]

Work out the sum of the interior angles of a decagon (10 sides).

.....

2 [2 marks]

A regular polygon has an interior angle of 162° .
How many sides does it have?

.....

3 [3 marks]

In a regular polygon, each interior angle is 4 times the size of each exterior angle.
How many sides does the polygon have?

.....

4 [3 marks]

The angles of a triangle are $(3x - 10)^\circ$, $(2x + 20)^\circ$ and $(x + 50)^\circ$.
Work out the size of each angle.

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5 [4 marks]

Prove that the exterior angle of a triangle is equal to the sum of the two interior angles at the other two vertices.

.....

6 [4 marks]

A regular hexagon, a square and one other regular polygon fit together exactly around a point, with no gaps.
How many sides does the third polygon have?

.....

7

[4 marks]

A regular octagon and a regular pentagon are placed so that they share one complete side. Work out the size of the angle between them at one of the shared vertices, measured outside both shapes.

8

[4 marks]

The five interior angles of a pentagon are x° , $(x + 20)^\circ$, $(x + 30)^\circ$, $(x + 40)^\circ$ and $(x + 50)^\circ$. Work out the size of the largest angle.

9

[4 marks]

Prove, using the properties of parallel lines, that the angles of a triangle add up to 180° .

10

[5 marks]

In a regular polygon, each interior angle is 20° more than 7 times each exterior angle.

- (a) How many sides does the polygon have?
 - (b) Work out the size of one interior angle.
 - (c) Work out the sum of the interior angles.
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